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Shrinkage Estimation in Finance and Insurance: Portfolio Construction and Claim Severity Modelling

SMM069 (2025–26) Group Coursework – Group 12

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Coursework 1

A Shrinkage estimation for portfolio construction

A.1 Data preprocessing

The risk-free rate was first converted from a percentage to a decimal (i) before weekly excess returns were computed (ii) using Eq. (1).

$$r_{i,t}^{\text{excess}} = r_{i,t} - r_{f,t} \quad (1)$$

The risk-free rate was aligned to the return dates, and the excess returns were calculated using the `sweep()` function to subtract the risk-free vector across the rows. The resulting excess return matrix covers 1251 weekly observations for 100 companies, from 2000-01-14 to 2023-12-29. In **R**, both of these steps were implemented using:

```
## (i) convert risk-free rate from percentage -> decimal ----
rf_dec    <- rf_raw
rf_dec$RF <- rf_raw$RF / 100
# align risk free rate to the return dates by date
dat <- merge(ret_raw, rf_dec[, c("Date", "RF")], by = "Date")
dates <- dat$Date
# original tickers, kept for GICS
ticker_cols <- setdiff(colnames(ret_raw), "Date")
returns_all <- as.matrix(dat[, ticker_cols])
rf <- dat$RF
# gets our specific tickers
set.seed(group_num)
selected <- sample(ticker_cols, no_companies)
returns100 <- returns_all[, selected]

## (ii) Weekly excess returns: r_excess[t,i] = r[t,i] - rf[t] ----
# sweep() subtracts the rf vector across rows (clearer than returns100 - rf)
excess_returns <- sweep(returns100, 1, rf, FUN = "-")
```

A.2 Selection of estimation models

Portfolios were constructed using all eight mean-vector estimators: the sample mean, the four distribution-free shrinkage estimators of Asimit et al. [2026] (`St-MSh`, `D-MSh`, `O-LSh` and `T-LSh`), and the three established benchmarks (`Wang`, `BOP` and `Jorion`). Please see Appendix 1.1 for formal definitions of each estimator.

Every mean estimator is paired with the one-parameter linear shrinkage estimator of Ledoit and Wolf [2004], `cov1Para`, rather than the sample covariance. `cov1Para` pulls the sample covariance S toward a diagonal matrix μI whose common variance $\mu = \frac{1}{p} \text{tr}(S)$ is the average sample variance. The strength of the pull is set automatically, with no tuning parameter, to minimise the covariance’s mean-squared error (Appendix 1.1). The result is always positive-definite and well-conditioned. It is also distribution-free, needing only i.i.d. observations with finite fourth moments, the same assumptions that underlie the mean estimators of Asimit et al. [2026]. The whole pipeline therefore rests on one consistent set of assumptions.

The issue with the sample covariance is conditioning. Each training window gives $n = 156$ weekly observations on $p = 100$ assets, a ratio of $n/p \approx 1.56$. Because $n > p$ the sample covariance can be inverted, but it is ill-conditioned. Its smallest eigenvalues are biased toward zero, so inverting the covariance amplifies the directions estimated least reliably [Ledoit and Wolf, 2004], producing extreme, unstable weights whose errors accumulate out of sample.

Both the problem and the fix are visible directly in our data. Averaged across the 273 windows (described in more detail in A.3), the condition number of the sample covariance is 9,812 compared to 221 under `cov1Para` – a 44-fold improvement (medians 5,820 vs 176). With the mean estimators and portfolio rule held fixed, the better conditioning feeds straight into implementability. For the `O-LS` tangency portfolio, average gross leverage falls from 47.0 to 7.6 and monthly turnover from 63.5 to 5.0, with comparable reductions for every estimator (Appendix 1.3).

A.3 Portfolio construction and rolling-window setup

For each window and each mean estimator, two portfolios are built: a mean-variance portfolio with risk aversion $\gamma = 1$, giving the unnormalised weights w_{MV} , and a tangency portfolio, the same vector rescaled to sum to one, w_{TP}

$$w_{MV} = \Sigma^{-1} \hat{\mu}, \quad w_{TP} = \frac{\Sigma^{-1} \hat{\mu}}{\mathbf{1}^\top \Sigma^{-1} \hat{\mu}}$$

The equal-weight portfolio (EWP) is included as a benchmark. In our code, γ is denoted `omega`. Throughout, Σ is the `cov1Para` estimate selected in A.2, and $\hat{\mu}$ is estimated on the training window only, so no test-window information enters the weights.

The rolling structure uses a 156-week (3-year) training window, a 4-week (1-month) testing window, and a 4-week step. Because the step equals the testing length, successive test windows are contiguous and non-overlapping. Over the 1251-week sample this gives

$$\lfloor (1251 - 156 - 4)/4 \rfloor + 1 = 273$$

out-of-sample windows, leaving the final 3 weeks uncovered by any test window. The implementation of this rolling structure in **R** was:

```

# build mean variance and tangency portfolio weights per estimator (& EWP)
# > Score each by annualised Sharpe ratio over 4 week test window
# > Then roll forward across sample (described in labs)
eval_window <- function(train_X, test_X, cov_fun, mean_estimators, omega = 1) {
  Sigma_inv <- solve(cov_fun(train_X))
  out <- numeric(0)
  for (nm in names(mean_estimators)) {
    mu <- mean_estimators[[nm]](train_X)
    # Sigma^{-1} mu
    z <- as.vector(Sigma_inv %*% mu)
    # mean-variance (gamma = 1) <-> un-normalised
    w_mv <- (1 / omega) * z
    # tangency -> weights sum to 1
    w_tp <- z / sum(z)
    out[paste0("MV_", nm)] <- ann_sharpe(test_X %*% w_mv)
    out[paste0("TP_", nm)] <- ann_sharpe(test_X %*% w_tp)
  }
  w_ew <- rep(1 / ncol(train_X), ncol(train_X))
  # equal-weight benchmark
  out["EWP"] <- ann_sharpe(test_X %*% w_ew)
  out
}

sr_list <- vector("list", n_windows)
test_start_dates <- as.Date(rep(NA, n_windows))
for (k in seq_len(n_windows)) {
  tr_start <- 1 + (k - 1) * step_weeks
  tr_end <- tr_start + training_weeks - 1
  te_start <- tr_end + 1
  te_end <- te_start + testing_weeks - 1
  train_X <- excess_returns[tr_start:tr_end, , drop = FALSE]
  test_X <- excess_returns[te_start:te_end, , drop = FALSE]
  sr_list[[k]] <- eval_window(train_X, test_X,
    cov_fun      = cov_fun,
    mean_estimators = mean_estimators,
    omega        = omega
  )
  test_start_dates[k] <- dates[te_start]
}
sr_mat <- do.call(rbind, sr_list)
sr_df <- data.frame(TestStart = test_start_dates, sr_mat, check.names = FALSE)

```

A.4 Portfolio performance

For every test window and every portfolio the annualised Sharpe ratio was calculated by taking the mean of the four weekly out-of-sample excess returns, dividing by their standard deviation and multiplying by $\sqrt{52}$. Each window is assigned to a market environment by its test-window start date. Of the 273 windows, 37 fall in the financial-crisis sub-period (06/10/2008–03/08/2011) and 49 in the pandemic sub-period (26/02/2020–29/12/2023). These boundary dates are structural breakpoints identified by the Bai-Perron test on the S&P500 return series and the sub-periods correspond to statistically identified market regimes rather than arbitrary calendar choices.

Table 1 shows the mean and standard deviation of the per-window Sharpe ratios over the full period and the two sub-periods. Each per-window Sharpe ratio is estimated from only four weekly returns, so it is inherently noisy. Cross-window standard deviations range from approximately 5.1 to 9.4 over the full period, in every case several times the magnitude of the corresponding mean (approximately 0.20 to 1.43). This dispersion is a property of the per-window definition. A companion summary for a Sharpe ratio computed on the stitched out-of-sample series is described in Appendix 1.4.

Note that the Sample and `St-MSh` rows in Table 1 are identical in every period. This is a structural property of the Sharpe ratio. `St-MSh` scales the sample mean by a positive constant that the Sharpe ratio cannot detect. This is explored further in Appendix 1.2 (the same identity is also present in Table 3 of Asimit et al. [2026]).

A.5 Interpretation of results and final report

Optimised portfolios are only as reliable as the expected returns they are built on, and those estimates carry substantial estimation error. This report tests whether shrinkage methods designed to temper that noise deliver better risk-adjusted returns than the classical sample mean for an investor rebalancing monthly over two decades [Asimit et al., 2026]. Using weekly excess returns for 100 S&P 500 companies (January 2000–December 2023), eight ways of estimating expected returns were compared, each paired with the `cov1Para` covariance estimator [Ledoit and Wolf, 2004] and benchmarked against an equal-weight (1/N) portfolio. Every method was evaluated with a rolling window that mirrors live practice, fitting on the previous three years, testing on the next unseen month, then advancing, giving 273 out-of-sample tests with no look-ahead bias. Performance was summarised by the annualised Sharpe ratio over the full period, the 2008 financial crisis, and the COVID-19 pandemic (Table 1).

The mean-variance and tangency portfolios hold the same positions at different scales (tangency rescales the weights to sum to one) so within any window their Sharpe ratios are identical bar a possible sign flip (the per-window magnitudes agree to within $4.4\text{e-}13$). The two coincide in every window where the portfolio is not net-short, which covers almost the entire pandemic. Over other periods the sign flip lifts the tangency full-period mean above mean-variance for the stronger estimators (`TP_OLSh` 1.1271 versus `MV_OLSh` 0.7252). The economically meaningful difference is cost, not Sharpe ratio. Because tangency weights sum to one, leverage and turnover are far lower (for `O-LSh`, average leverage falls from 88.5 to 7.6), making tangency preferable in practice. The exception is when the nor-

Table 1: Mean and standard deviation of the per-window annualised Sharpe ratios. The best mean and standard deviation in each environment are highlighted. Because the Sharpe ratio is invariant to positive rescaling of the weights, St-MSh reproduces the Sample rows exactly, and the mean-variance and tangency portfolios coincide except in net-short windows where the sign flips.

Strategy	Full period		Financial crisis		Pandemic	
	Mean	SD	Mean	SD	Mean	SD
Benchmark						
EWP	1.4344	5.0685	2.0548	5.6329	0.9005	4.7741
Mean-variance portfolio						
StMSh	0.6131	5.7726	-0.1796	6.9569	0.2372	4.3007
DMSH	0.8917	6.1426	0.7120	6.5157	0.5848	4.9326
OLSh	0.7252	5.5790	0.1695	6.6529	0.0543	4.6813
TLSh	0.6411	5.5038	-0.3894	6.9169	0.2494	4.5395
Sample	0.6131	5.7726	-0.1796	6.9569	0.2372	4.3007
Wang	0.3845	5.1666	1.2123	7.1428	-0.1392	5.4192
BOP	0.1961	5.9551	0.0162	6.5120	-0.0583	5.4877
Jorion	0.3925	9.4441	-0.4206	6.8191	-2.2950	17.9302
Tangency portfolio						
StMSh	0.5801	5.7760	-0.0174	6.9592	0.2372	4.3007
DMSH	1.1108	6.1066	1.0847	6.4627	0.5848	4.9326
OLSh	1.1271	5.5116	2.3099	6.2296	0.0543	4.6813
TLSh	0.3455	5.5303	-2.1559	6.5744	0.2494	4.5395
Sample	0.5801	5.7760	-0.0174	6.9592	0.2372	4.3007
Wang	0.4908	5.1575	0.8808	7.1925	-0.0698	5.4206
BOP	0.2399	5.9534	-0.9459	6.4410	-0.0848	5.4873
Jorion	0.7141	9.4252	1.4786	6.6659	-2.2950	17.9302

maliser nears zero and weights explode (30 January 2009 drives TP_DMSH leverage to roughly 492,421; Appendix 1.6).

Across estimators, the standout result is how hard the naïve benchmark is to beat. The equal-weight portfolio posts the highest full-period Sharpe (1.4344) and lowest full-period dispersion, overtaken only once, by TP_OLSh in the financial crisis (2.3099 versus 2.0548), consistent with the 1/N result of DeMiguel et al. [2009]. Among optimised methods only mean shrinkage keeps up. Under the tangency construction, O-LSH and D-MSh roughly double the sample mean's full-period Sharpe ratio (1.1271 and 1.1108 versus 0.5801), with D-MSh the most consistent across regimes and the best optimised pandemic performer (0.5848). The raw sample mean turns negative in the financial crisis (-0.1796) through error-maximisation, St-MSh reproduces the sample rows exactly (Appendix 1.2), Jorion is too volatile to trust (SD 9.44, rising to 17.93 in the pandemic), and BOP is the weakest overall, with the most net-short flips (62 windows). The uniformly weak pandemic figures are partly due to our group's sector draw (Figure 1; full company list in Appendix 1.7): Financials is the largest single exposure (14 of 100), and while Information Technology sits mid-pack (12), the tech-led sectors that drove the post-2020 rally are thin as a bloc with IT and Communication Services (4) together only accounting for 16% of the universe. No estimator can manufacture exposure that isn't there and an equal-weighted portfolio sidesteps this problem by never guessing. For our 100 stocks, O-LSH and D-

MSh are worth using over the raw sample mean, but once leverage and turnover are counted, a plain equal-weighted portfolio remains a strong, cheap benchmark the optimisers struggle to beat out of sample (Appendix 1.4).

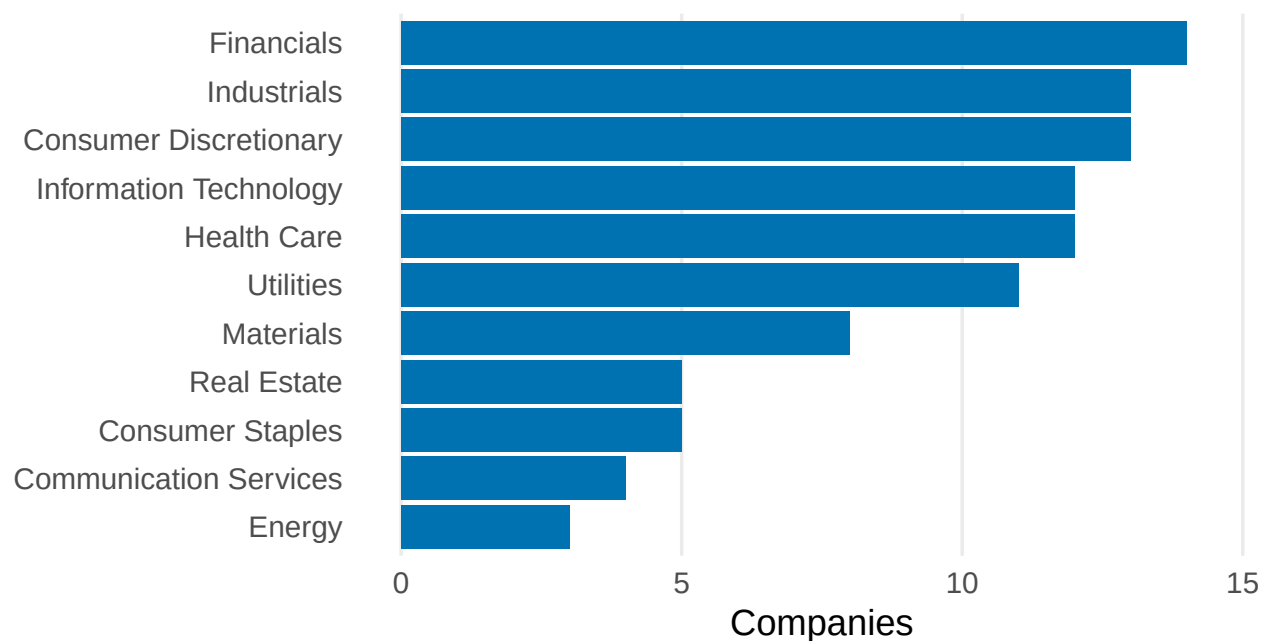


Figure 1: GICS sector composition of the 100 sampled companies (Group 12), shown as a horizontal bar chart with sectors ordered by company count. Financials is the largest exposure, while Communication Services and the wider tech-led bloc are comparatively underweight.

Coursework 2

B Shrinkage Generalised Linear Models for Insurance Claim Severity Estimation

B.1 Out-of-sample performance evaluation

B.1.i Data and preprocessing

The French Motor Third-Party Liability (freMTPL2) frequency and severity datasets were merged on policy ID after aggregating claim amounts per policy. Policies with exactly one claim and a positive claim amount were retained, and a random sample of 1,000 policies was drawn under our Group 12 seed. This sample was fixed for all subsequent analysis, with the cross-validation seeds inside the evaluation loop reshuffling only the fold labels.

Three groups of features were then constructed. Continuous rating factors were capped and log-transformed, with `VehPower` capped at 9, `BonusMalus` at 150 and `Density` floored at 1, to bound implausible values and squash the heavy right tails typical of insurance covariates. Driver and vehicle age entered through natural cubic splines with three degrees of freedom, allowing the non-monotone age effects familiar from motor pricing. The high-cardinality factors `Region`, `Area` and `VehBrand` were target-encoded with a credibility-style smoother,

$$\text{TE} = \lambda \bar{y}_{\text{group}} + (1 - \lambda) \bar{y}_{\text{global}}$$

with $\lambda = n / (n + \kappa)$ and $\kappa = 20$, computed leave-one-out on training rows so that a policy never contributes to its own encoding. Target encoding also keeps the design matrix compact. One-hot encoding these three factors would add roughly thirty sparse columns to an 800-row training fold, worsening the conditioning of the weighted information matrix that must be inverted at every step of the iteratively reweighted least squares (IRLS) algorithm.

All continuous features were standardised using training-fold means and standard deviations. Standardisation is not strictly required for any of the fitted models, since fitted values depend on the data only through the linear predictor, but it is retained for comparability with the released practice code. Every preprocessing step was learned on training folds and frozen before application to test folds, so no information leaks from test to train. The robustness of these choices, covering the two variable caps, the spline flexibility, the target-encoding smoother κ and the standardisation step, is examined in Appendix 2.1. In **R** this was implemented as follows:

```
# Read in data (stored in relative data/ folder)
freq <- read.csv("data/freMTPL2freq.csv")
sev <- read.csv("data/freMTPL2sev.csv")
```

```

# Aggregate severity by policy ID (& merge with frequency features)
sev_agg <- aggregate(ClaimAmount ~ IDpol, data = sev, FUN = sum)
data <- merge(freq, sev_agg, by = "IDpol")
# Filter for exactly 1 claim and positive claim amount
data_sev <- data[data$ClaimNb == 1 & data$ClaimAmount > 0, ]

# Group seed set before the single random draw of 1,000 rows just to be safe
# (avoiding part A rng leakage)
set.seed(group_num)
# Randomly sample 1000 rows
data_sev <- data_sev[sample(nrow(data_sev), 1000), ]
# for generality (= ClaimAmount)
data_sev$Severity <- data_sev$ClaimAmount / data_sev$ClaimNb

# Squash heavy tails and bound implausible values.
# > defaults:
#   VehPower capped at 9
#   BonusMalus at 150
# sensitivity to both caps is tested in the appendix (apx-b-caps)
data_sev$VehPower_Log <- log(pmax(pmin(data_sev$VehPower, 9), 1))
data_sev$Density_Log <- log(pmax(data_sev$Density, 1))
data_sev$BonusMalus_Log <- log(pmin(data_sev$BonusMalus, 150))
# Quick check -> should be 1000
n_pol <- nrow(data_sev)

```

B.1.ii Models and out-of-sample evaluation design

Nine models were fitted with a Gamma response and log link such that

$$\hat{y}_i = \hat{\mu}_i = \exp(x_i^\top \hat{\beta})$$

The IRLS fit `GLM2` serves as the baseline. Ridge regression (RR) and elastic net (EN) were fitted via `cv.glmnet`, with the penalty chosen by internal cross-validation (`lambda.min`). The remaining six models are the shrinkage GLMs of Asimit et al. [2025] implemented in the `savvyGLM` package (`St`, `DSh`, `SR`, `GSR`, `LW` and `QIS`). These estimators shrink the maximum-likelihood estimates towards structured targets in the spirit of the mean and covariance shrinkage used in Coursework 1. Mechanically, these estimators retain the IRLS structure of `glm.fit2` but replace the weighted least squares (WLS) step at each iteration with a shrunk solution. `St` and `DSh` rescale the WLS coefficients by scalars chosen to minimise mean squared error, `SR` and `GSR` impose slab penalties along eigenvector directions, and `LW` and `QIS` shrink the weighted information matrix before inversion. All shrinkage intensities are available in closed form, so unlike RR and EN no cross-validation is required. RR and EN are fitted by coordinate descent, which is why their computation time (CT) and iteration count (NoIt) are not

directly comparable with those of the IRLS family.

Performance in each run is scored by the metrics specified in the brief. The root mean squared error (RMSE) and mean absolute error (MAE) measure raw prediction error, with winsorised variants capping actual and predicted values at the 99th and 95th training-response percentiles (W99, W95) to limit distortion from extreme claims. The Gamma deviance (Dev) measures goodness-of-fit under the assumed distribution, the raw and normalised Gini indices measure risk-ranking ability, and the actual-to-expected (AE) ratio measures aggregate calibration, with a value of one indicating that total predicted severity matches total observed severity.

Out-of-sample performance is estimated by 5-fold cross-validation replicated 100 times, giving 500 evaluation runs. In each replication the 1,000 policies are partitioned into five folds, with 80% of observations training the models and 20% testing them, and each fold serves as the test set exactly once. Winsorisation caps for the robust error metrics are computed from the training response in each run. The evaluation loop is reproduced below, with results cached to an rds file and reloaded at knit time under the final configuration (one-claim filter, group seed 12, $n = 1,000$).

```
# NOTE: the full 100 x 5 = 500-run loop takes ~20 minutes
# results are pre-computed once and cached to final_results.rds (loaded below)
n_repeats <- 100
n_iter <- n_repeats * n_folds
results_list <- vector("list", n_iter)
for (i in seq_len(n_iter)) {
  results_list[[i]] <- run_single_iteration(i)
  cat(sprintf("Finished iteration %d of %d.\n", i, n_iter))
}
final_results <- do.call(rbind, results_list)
saveRDS(final_results, "data/final_results.rds")
```

Convergence warnings emitted by `glmnet` and by the `savvyGLM` step-halving safeguard during the loop are expected behaviour of those fitters rather than evidence of failure, and are discussed further in Appendix 2.2.

B.1.iii Raw performance

Table 2 reports each metric averaged over all 500 runs. Because severity is heavy-tailed, the unwinsorised RMSE is dominated by a handful of very large claims and separates the models poorly, so the winsorised variants are the more informative measures of accuracy. A second feature of the table shapes the whole comparison. Every normalised Gini is negative, baseline included (the least negative is EN at -0.039), whereas a model with genuine ranking power has a normalised Gini in $(0, 1]$. At $n = 1,000$, policy-level claim severity therefore carries essentially no rankable out-of-sample signal. This is an expected property of individual severity, where most variation is claim-level randomness, rather than a defect of any particular estimator. The Gini columns are retained for completeness, and the comparison between models rests on the winsorised errors, the Gamma deviance and the AE calibration ratio.

Table 2: Raw out-of-sample performance: mean of each metric over all 500 evaluation runs. The best value per column is shown in green (lower is better for the error, deviance, time and iteration metrics, higher for the Gini indices, and closest to one for the AE ratio). Nolt counts IRLS iterations for GLM2 and savvyGLM but coordinate-descent passes for glmnet, so it is not comparable across those two families.

Model	RMSE	MAE	RMSE W99	RMSE W95	MAE W99	MAE W95	Dev.	Gini raw	Gini norm	AE	CT (s)	Nolt
Baseline												
GLM2	6888	1861	2378	1437	1380	1135	369.6	-0.0697	-0.1025	1.024	0.005	13.3
Penalised (glmnet)												
RR	6186	1774	2232	1301	1354	1123	321.5	-0.0324	-0.0448	1.033	0.608	781.1
EN	6197	1781	2239	1310	1361	1129	325.4	-0.0278	-0.0388	1.029	0.510	1130.9
Shrinkage GLMs (savvyGLM)												
QIS	7323	1826	2353	1373	1309	1064	389.4	-0.0958	-0.1453	1.072	0.020	12.8
LW	8040	2130	2520	1632	1565	1311	414.3	-0.1217	-0.1913	0.907	0.023	11.6
SR	6402	1926	2483	1574	1500	1234	417.0	-0.0813	-0.1189	0.993	0.028	6.3
GSR	7263	1887	2386	1441	1377	1131	365.0	-0.0685	-0.1033	1.025	0.029	7.8
St	6913	1827	2363	1406	1343	1098	371.5	-0.0698	-0.1027	1.052	0.026	11.3
DSh	7119	1892	2356	1403	1389	1144	344.9	-0.0655	-0.1000	0.997	0.030	9.7

Table 3: Relative out-of-sample performance: mean of per-run ratios versus GLM2 (GLM2 = 1 by construction). Values below one are better for the error, deviance, time and iteration metrics. The AE ratio is itself a ratio and the per-run GLM2 Gini is near zero and sign-varying, so ratios of these three columns are not meaningful. As they are judged from the raw table their cells are omitted here.

Model	RMSE	MAE	RMSE W99	RMSE W95	MAE W99	MAE W95	Dev.	Gini raw	Gini norm	AE	CT (s)	Nolt
Baseline												
GLM2	1.000	1.000	1.000	1.000	1.000	1.000	1.000	–	–	–	1.000	1.000
Penalised (glmnet)												
RR	0.918	0.963	0.940	0.908	0.983	0.990	0.899	–	–	–	129.866	60.849
EN	0.919	0.967	0.943	0.913	0.988	0.996	0.908	–	–	–	106.127	82.992
Shrinkage GLMs (savvyGLM)												
QIS	1.047	0.973	0.989	0.957	0.949	0.939	1.031	–	–	–	4.551	0.966
LW	1.152	1.143	1.062	1.138	1.136	1.157	1.104	–	–	–	5.022	0.968
SR	0.961	1.041	1.043	1.095	1.086	1.087	1.103	–	–	–	6.137	0.515
GSR	1.040	1.010	1.004	1.002	0.998	0.997	0.994	–	–	–	6.323	0.636
St	1.000	0.979	0.993	0.979	0.973	0.968	1.002	–	–	–	5.709	0.869
DSh	1.033	1.017	0.991	0.977	1.007	1.009	0.953	–	–	–	6.459	0.748

B.1.iv Relative performance

Table 3 shows the average of the per-run ratios against the GLM2 value in the same run for each metric, given by Eq. (2).

$$\frac{1}{500} \sum_{j=1}^{500} \frac{\text{Metric for Model}^{(j)}}{\text{Metric for GLM2}^{(j)}} \quad (2)$$

Because each ratio conditions on the same train/test split, fold-to-fold noise cancels and the table isolates the systematic effect of the estimator. Three columns are excluded from ratio form. The AE ratio is already a ratio, so a further ratio has no calibration interpretation. The same exclusion applies with greater force to the two Gini indices, whose per-run GLM2 denominator is small and frequently negative, so the resulting ratios flip sign and explode. The mechanism is the same division by a near-zero quantity that inflated the tangency leverage in Coursework 1 (Appendix 1.6), with one important difference. In Coursework 1 the denominator was part of the portfolio construction, so the explosion was a genuine economic property of the strategy. Here it is an artefact of the reporting convention and car-

ries no information about the models. All three metrics are therefore judged from the raw table (Table 2) and their ratio cells are omitted.

Where the ratios are interpretable, a consistent ordering emerges. RR delivers the largest systematic reductions in the winsorised RMSEs (9.2% at the 95% cap) and in deviance, with EN a near-tie, while the `savvyGLM` estimators cluster around parity with `GLM2` at a fraction of `glmnet`’s computation time. This ordering is consistent with the evidence in Asimit et al. [2025], which finds ridge regression dominant under the log link used here in both simulation and the `freMTPL2` application.

B.2 Model comparison and selection

With no model showing out-of-sample ranking power (Section B.1.iii), selection rests on tail-robust accuracy, deviance and calibration. On that basis we select Ridge Regression (RR) as the “best” overall performer. RR posts the best winsorised RMSE at both caps and the lowest deviance, beating `GLM2` on the 95%-capped RMSE in 99.8% of the 500 runs with a 9.2% average reduction. Elastic net is practically indistinguishable, within 0.6 percentage points on the 95%-capped RMSE with marginally better calibration, and would be an equally defensible choice. RR’s aggregate calibration (AE 1.033) trails DSh (0.997), but a uniform 3% shortfall is corrected by rebasing the premium level, whereas DSh cannot recover RR’s accuracy. That penalised L2 beats the bespoke shrinkage GLMs is itself informative. At $n = 1,000$, with a correlated spline and target-encoded design and almost no severity signal, the optimal shrinkage is heavy. Ridge tunes its intensity directly against predictive loss by cross-validation, while the `savvyGLM` plug-in intensities shrink towards maximum-likelihood targets that inherit the noise, the regime in which Asimit et al. [2025] likewise find ridge dominant under the log link. RR’s computation time, under a second per fit, is immaterial.

B.3 In-sample performance analysis

The chosen model, RR, was refitted to the full sample of 1,000 policies. The preprocessing described in Section B.1.i was applied unchanged, with the splines, leave-one-out target encoding and standardisation now learned from all 1,000 rows and the penalty re-selected by `cv.glmnet` under our group seed (12) so that the fit was stable between knits. Its fitted severities $\hat{y}_i$ are compared with the observed claims y_i below.

At the portfolio level the model is almost perfectly calibrated. The actual-to-expected ratio is 1.001, so total predicted severity matches total observed severity to within 0.1%. Beneath that aggregate, however, the model differentiates very little. The Gamma deviance falls only 0.2% below the intercept-only model, and the 1,000 fitted severities span a band of roughly 39 euros (€1,933 to €1,972) around the portfolio mean, against actual claims ranging from €12 to €182,500. Fitted to all 1,000 rows, `cv.glmnet` selects a heavy penalty and shrinks the model nearly to the intercept. This is the in-sample face of the selection argument in B.2. With almost no rankable severity signal, the loss-minimising fit sits close to a constant. RR won the out-of-sample comparison precisely because it declines to fit noise. Absolute errors remain large (RMSE 7,392, MAE 1,780), an unavoidable feature of individual-level severity, where most of the variance is claim-level randomness that is not predictable from any policyholder characteristic present in the data.

Table 4 places the in-sample metrics beside the unpenalised GLM2 baseline fitted to the same rows. GLM2 posts the lower in-sample RMSE, the larger deviance reduction and the higher normalised Gini, yet Table 2 showed it losing to RR on all the error metrics. The additional in-sample fit that GLM2 achieves does not generalise, which is the clearest demonstration of what the ridge penalty removes. GLM2 also miscalibrates in aggregate (AE 0.948), over-predicting total severity by 5.2%, so the penalty improves the portfolio level as well as the policy level.

Table 4: In-sample performance of the selected RR model and the unpenalised GLM2 baseline, both fitted to all 1,000 policies. GLM2 fits the sample more closely on every measure, but that additional fit does not survive out of sample.

Metric	RR	GLM2
RMSE (EUR)	7,392	8,999
MAE (EUR)	1,780	1,917
AE ratio	1.001	0.948
Normalised Gini	0.285	0.318
Deviance reduction vs intercept-only	0.2%	9.3%

Figure 2 illustrates these points visually by placing RR beside the unpenalised GLM2 fitted to the same data. GLM2 spreads its predictions across an order of magnitude while RR compresses them to a narrow vertical band. However, Table 2 shows that the wider spread carries no out-of-sample ranking power. It is fitted noise, and removing it is exactly what the penalty is for. Panel B plots the deviance residuals. With near-constant predictions these display the dispersion of individual claims around a single level, rather than acting as a diagnostic of the link specification. The cloud is centred on zero with the positive skew of the severity tail.

Table 5: In-sample calibration of RR by decile of predicted severity, with the decile actual-to-expected ratio.

Decile	N	Mean predicted (EUR)	Mean actual (EUR)	AE ratio
1	100	1942	1164	0.599
2	100	1946	1602	0.823
3	100	1948	1147	0.589
4	100	1950	1990	1.021
5	100	1951	2589	1.327
6	100	1953	1608	0.824
7	100	1954	1552	0.794
8	100	1956	1803	0.922
9	100	1959	1290	0.659
10	100	1964	4801	2.444

Table 5 sorts policies into deciles of predicted severity. Mean predicted severity is essentially flat, rising from €1,942 in the bottom decile to €1,964 in the top, while mean actual severity swings from €1,147 to €4,801. The decile AE ratios are patternless across the lower nine deciles (0.59 to 1.33) before jumping to 2.44 in the top decile. With a near-constant predictor, decile membership mostly slices noise, and the top decile happens to capture a cluster of large claims. That cluster is the sole source of the

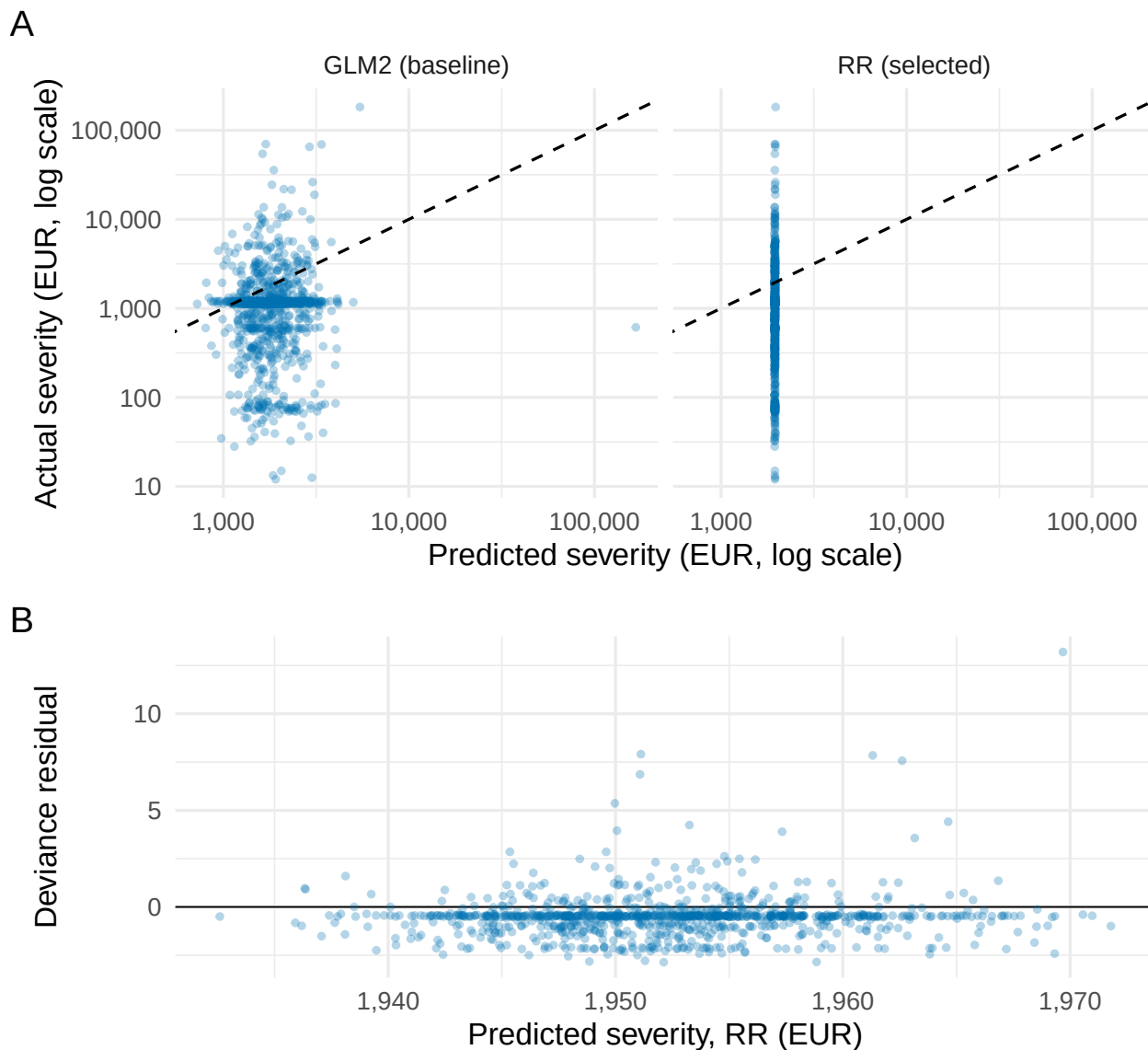


Figure 2: In-sample diagnostics for RR on the full sample. Panel A: fitted against actual severity (log scales) for the unpenalised GLM2 baseline and the selected RR model, with the dashed line marking perfect prediction. Panel B: RR deviance residuals against predicted severity.

modest in-sample normalised Gini of 0.285, a figure that fails to survive out of sample, where every model's Gini is negative.

At this volume, average claim severity is best rated near-flat. In the standard frequency-severity decomposition, risk differentiation is carried by the frequency model, and exposure to the largest claims is managed through large-claim loadings or layered reinsurance rather than the policy-level severity model. As such, RR's behaviour is not a defect but the coherent endpoint of the analysis. It is an almost perfectly calibrated aggregate level, and an honest refusal to manufacture differentiation the data cannot support.

Appendices

1 Coursework 1: Supplemental methodology

1.1 Mean-vector estimators

Every estimator starts from the sample mean vector, $\bar{X}$. The shrinkage estimators adjust $\bar{X}$ using the unbiased sample covariance of the returns, S , and the grand mean $\bar{X}_g = \frac{1}{p} \mathbf{1}^\top \bar{X}$, the average of $\bar{X}$ across the p assets. Each estimator maps the $n \times p$ matrix of training returns to its own length- p estimate, and the estimators are compared by mean squared error (MSE).

Table 6: Mean-vector estimators: shrinkage target, mechanism, and the loss each is designed to minimise.

Estimator	Shrinks toward	How it shrinks	Loss	Source
Sample mean	no shrinkage	Unbiased; high variance when $p \approx n$	—	—
Stein Multiplicative Shrinkage (St-MSh)	zero	One scalar $\hat{c} \in [0, 1)$ applied to every asset	MSE	Asimit et al. [2026]
Diagonal Multiplicative Shrinkage (D-MSh)	zero	Asset-specific scalars $\hat{c}_k$ (suits differing scales)	MSE	Asimit et al. [2026]
One-Parameter Linear Shrinkage (O-LSh)	grand mean	One intensity $\hat{\delta}$	MSE	Asimit et al. [2026]
Two-Parameter Linear Shrinkage (T-LSh)	grand mean, rescaled	Intensity $\hat{\delta}$ plus target scaling $\hat{\xi}$	MSE	Asimit et al. [2026]
Shrinkage estimator of Wang et al. (Wang)	grand mean	One intensity, estimated by U-statistics	MSE	Wang et al. [2014]
Shrinkage estimator of Bodnar et al. (BOP)	precision-weighted grand mean	One intensity, built from S^+	invariant	Bodnar et al. [2019]
Bayes–Stein estimator (Jorion)	precision-weighted grand mean	Bayes–Stein intensity	invariant	Jorion [1986]

The four estimators of Asimit et al. [2026] take the following explicit forms:

$$\begin{aligned} \hat{\mu}^{\text{St-MSh}} &= \hat{c} \bar{X}, & \hat{c} &= \frac{\|\bar{X}\|_p^2}{\|\bar{X}\|_p^2 + \frac{1}{np} \text{tr}(S)}, & \|\bar{X}\|_p^2 &= \frac{1}{p} \bar{X}^\top \bar{X}, \\ \hat{\mu}_k^{\text{D-MSh}} &= \hat{c}_k \bar{X}_k, & \hat{c}_k &= \frac{\bar{X}_k^2}{\bar{X}_k^2 + \frac{1}{n} S_{kk}}, \\ \hat{\mu}^{\text{O-LSh}} &= (1 - \hat{\delta}) \bar{X} + \hat{\delta} \bar{X}_g \mathbf{1}, & \hat{\mu}^{\text{T-LSh}} &= (1 - \hat{\delta}) \bar{X} + \hat{\delta} \hat{\xi} \bar{X}_g \mathbf{1}. \end{aligned}$$

The amount of shrinkage in each estimator is set by a shrinkage intensity, $\hat{c}$ or $\hat{c}_k$. This scalar multiplies the sample mean in [St-MSh](#) and [D-MSh](#), or the weight $\hat{\delta}$ that mixes the sample mean with the

target in `O-LSh` and `T-LSh`. All four intensities follow the same signal-to-noise trade-off. They shrink harder when the sample mean is noisier and less when the signal is stronger. `T-LSh` extends `O-LSh` by also scaling the target, and its oracle MSE (the MSE it would attain if the true mean and covariance were known) is strictly lower than that of `O-LSh`.

The other three estimators, `Wang`, `BOP` and `Jorion`, are established benchmarks. `Wang` shares the target of `O-LSh`, the grand mean, but estimates its intensity with U-statistics rather than the sample plug-in estimates used above [Wang et al., 2014]. `BOP` and `Jorion` instead shrink toward a precision-weighted grand mean, a minimum-variance combination formed with S^+ for `BOP` and $\tilde{S}^{-1}$ for `Jorion` [Bodnar et al., 2019, Jorion, 1986]. Both optimise the invariant loss rather than the MSE, weighting estimation errors by the precision matrix, the inverse covariance.

1.2 Why `St-MSh` reproduces the sample-mean Sharpe ratios exactly

The identical Sample and `St-MSh` rows in Table 1 are not a coincidence, nor a bug. The same identity appears in Table 3 of Asimit et al. [2026] where the `St-MSh` and Sample rows are identical in both the mean-variance and tangency panels (e.g. 0.274/1.352 in Panel A and 0.385/1.325 in Panel B), despite their different universe (441 stocks), frequency (daily) and window scheme.

This occurs because `St-MSh` multiplies the sample mean by a single scalar, $\hat{\mu}^{\text{St-MSh}} = \hat{c} \bar{X}$, and this scalar is strictly positive, since the numerator of $\hat{c}$ is a squared norm and the denominator adds a sum of sample variances. Because the portfolio map is linear in $\hat{\mu}$, the scalar passes straight through to the weights,

$$w_{MV}^{\text{St-MSh}} = \Sigma^{-1}(\hat{c} \bar{X}) = \hat{c} w_{MV}^{\text{Sample}}$$

and a portfolio's weekly returns scale with its weights, $r_t(\hat{c} w) = \hat{c} r_t(w)$. The Sharpe ratio is a ratio of a mean to a standard deviation, so any positive constant cancels:

$$\text{SR}(\hat{c} w) = \frac{\hat{c} \bar{r}}{\hat{c} s_r} \sqrt{52} = \text{SR}(w)$$

The two strategies therefore have the same Sharpe ratio in every window, hence the same mean and the same standard deviation across windows. For the tangency portfolio the cancellation happens one step earlier, in the weights themselves,

$$w_{TP}^{\text{St-MSh}} = \frac{\hat{c} \Sigma^{-1} \bar{X}}{\hat{c} \mathbf{1}^\top \Sigma^{-1} \bar{X}} = w_{TP}^{\text{Sample}}$$

As a result `TP_StMSh` and `TP_Sample` are the same portfolio with identical leverage, turnover and net-short windows, as seen in Tables 7 and 8. Since $\hat{c} > 0$, `St-MSh` also flips sign in exactly the same windows as the sample mean.

An interesting observation is that because the scalar $\hat{c}$ is re-estimated each window the stitched `MV_StMSh` series is not a global rescaling of `MV_Sample` as each month is scaled by a different constant. The per-window Sharpe statistics are identical, but the stitched Sharpe in Table 8 differs slightly, and every scale-sensitive metric (leverage, volatility, certainty-equivalent) shrinks by roughly $\hat{c}$ for

the mean-variance portfolio. The other estimators avoid this equivalence because they change the direction. `D-MSh` rescales each asset differently, and the linear-shrinkage estimators mix $\bar{X}$ with a nonzero target. `St-MSh`'s shrinkage is visible in the scale-sensitive mean-variance metrics, but it is invisible to any scale-invariant evaluation of a linear portfolio rule.

1.3 Covariance conditioning: `cov1Para` vs sample covariance

With the mean estimators and portfolio construction held fixed, replacing `cov1Para` by the sample covariance sharply worsens conditioning and implementability. Averaged across windows, the condition number of the covariance estimate is 220.7 under `cov1Para` against 9,812.1 under the sample covariance, with medians of 175.5 and 5,819.6. Table 7 shows, for each mean estimator, the average gross leverage and turnover of the tangency portfolio under each covariance estimator, along with its Sharpe ratio on the stitched out-of-sample series.

Table 7: Tangency-portfolio implementability by covariance estimator. Gross leverage and turnover are averaged across windows, and Sharpe is computed on the stitched out-of-sample series.

	cov1Para			Sample covariance		
	Gross lev.	Turnover	Sharpe	Gross lev.	Turnover	Sharpe
StMSh	30.76	32.88	-0.41	139.63	190.90	0.01
DMSH	1836.89	3647.22	0.42	116.09	146.34	0.32
OLSh	7.62	5.03	0.42	46.95	63.49	0.04
TLSh	15.31	12.38	-0.26	207.26	360.01	0.19
Sample	30.76	32.88	-0.41	139.63	190.90	0.01
Wang	4.54	1.97	-0.01	12.67	6.67	-0.25
BOP	56.24	92.18	-0.12	110.75	143.05	-0.13
Jorion	11.72	12.64	0.29	53.98	74.99	-0.03

1.4 Scale-sensitive out-of-sample metrics

The annualised Sharpe ratio is invariant to positive rescaling of the weight vector, so it cannot distinguish portfolios that differ sharply in leverage, trading and exposure. However these are the features that determine whether a strategy can be run. Table 8 shows metrics that respond to scale computed on the stitched out-of-sample series, with the 273 test windows concatenated in time order. As such, the Sharpe ratios differ from the per-window average shown in Table 1.

The certainty-equivalent (CE) return is the risk-adjusted return an investor with risk aversion γ would trade the strategy for, computed as the annualised excess return minus $\frac{\gamma}{2}$ times the annualised variance. This uses the same $\gamma = 1$ as the portfolio construction (Section A.3). Gross leverage (the sum of absolute weights) and maximum absolute weight are averaged across windows. Turnover is the change in weights per monthly rebalance. The net-short count is the number of windows in which the tangency normaliser $c = \mathbf{1}^\top \Sigma^{-1} \hat{\mu}$ is negative.

Turnover compares target weights at consecutive rebalances and does not account for the drift of holdings by intervening returns. This approximates true trading volume and because the same simplifica-

tion applies to every strategy, cross-strategy comparisons are unaffected.

Table 8: Scale-sensitive out-of-sample metrics over the full period, using annualised weekly excess returns with risk aversion $\gamma = 1$. The Sharpe ratio is computed on the stitched out-of-sample series. The certainty-equivalent return is the annualised excess return minus half the annualised variance. Gross leverage (sum of absolute weights) and maximum absolute weight are averaged across windows. Turnover is measured per monthly rebalance. The net-short count is the number of windows with a negative tangency normaliser. Within each column, the best and second-best values are shown in green and orange respectively.

Strategy	Performance				Implementability			
	Ann. ex. ret.	Ann. vol.	Sharpe	Cert. equiv.	Gross lev.	Max wt	Turnover	Net-short
Benchmark								
EWP	0.107	0.202	0.527	0.1	1.0	0.01	0.0	–
Mean-variance portfolio								
StMSh	-0.501	4.571	-0.110	-10.9	88.2	3.25	28.4	–
DMSH	-0.736	6.346	-0.116	-20.9	125.0	5.27	43.8	–
OLSh	-0.366	4.850	-0.075	-12.1	88.5	3.22	27.5	–
TLSh	-0.460	4.561	-0.101	-10.9	82.8	3.04	26.7	–
Sample	-1.458	8.803	-0.166	-40.2	163.6	5.99	50.6	–
Wang	0.799	4.520	0.177	-9.4	75.1	2.60	26.1	–
BOP	1.925	8.628	0.223	-35.3	130.0	4.70	48.4	–
Jorion	-0.260	3.589	-0.072	-6.7	67.1	2.48	21.4	–
Tangency portfolio								
StMSh	-3.476	8.386	-0.414	-38.6	30.8	1.10	32.9	13
DMSH	1,626.349	3,878.088	0.419	-7,518,158.3	1,836.9	80.41	3,647.2	16
OLSh	0.418	0.992	0.421	-0.1	7.6	0.27	5.0	15
TLSh	-0.573	2.238	-0.256	-3.1	15.3	0.57	12.4	31
Sample	-3.476	8.386	-0.414	-38.6	30.8	1.10	32.9	13
Wang	-0.005	0.530	-0.010	-0.1	4.5	0.16	2.0	13
BOP	-1.708	13.687	-0.125	-95.4	56.2	2.33	92.2	62
Jorion	1.030	3.506	0.294	-5.1	11.7	0.43	12.6	23

1.5 Tangency sign flips

The tangency weights are $\Sigma^{-1}\hat{\mu}/c$ with $c = \mathbf{1}^\top \Sigma^{-1}\hat{\mu}$. When c is negative the construction returns the minimum-Sharpe portfolio rather than the maximum-Sharpe one, so the in-sample Sharpe ratio becomes large and negative. Counting these net-short windows is a useful diagnostic, because they tend to cluster in high-volatility regimes, exactly where the equal-weighted portfolio does well by never optimising. Table 9 shows the share of windows that flip and the average out-of-sample Sharpe in flipped versus non-flipped windows, for each estimator.

Figure 3 plots in-sample against out-of-sample Sharpe, with net-short windows highlighted. Out-of-sample Sharpe sits in a narrow band around zero across the full in-sample range, with no upward trend, showing that a strong in-sample Sharpe does not translate into a strong out-of-sample one. The flip windows separate the estimators by design. For the mean-shrinkage estimators (St-MSh, D-MSh, O-LSh, T-LSh) and Jorion, the flips concentrate at negative in-sample Sharpe, so these methods turn net-short mainly when recent performance was poor. Wang and BOP behave in the opposite way, flipping at positive in-sample Sharpe, and BOP flips far more often than any other estimator (22.7% of windows against 4.8% for Wang). This split aligns with the loss functions for each estimator (Appendix

1.1). Estimators that shrink toward the grand mean under mean-squared error flip in one regime, and the two that shrink toward the unit vector under the invariant loss flip in the other.

As before the Sample and `St-MSh` rows coincide exactly. As $\hat{c} > 0$, the sign of $c = \mathbf{1}^\top \Sigma^{-1} \hat{\mu}$ is unchanged by the shrinkage, so the two estimators flip in the same windows (Appendix 1.2).

Table 9: Tangency sign-flip diagnostic by estimator: share of net-short windows and average out-of-sample Sharpe in flipped versus non-flipped windows.

Estimator	Windows flipped (%)	OOS Sharpe (flipped)	OOS Sharpe (not flipped)
StMSh	4.76	-0.35	0.63
DMSH	5.86	1.87	1.06
OLSh	5.49	3.66	0.98
TLSh	11.36	-1.30	0.56
Sample	4.76	-0.35	0.63
Wang	4.76	1.12	0.46
BOP	22.71	0.10	0.28
Jorion	8.42	1.91	0.60

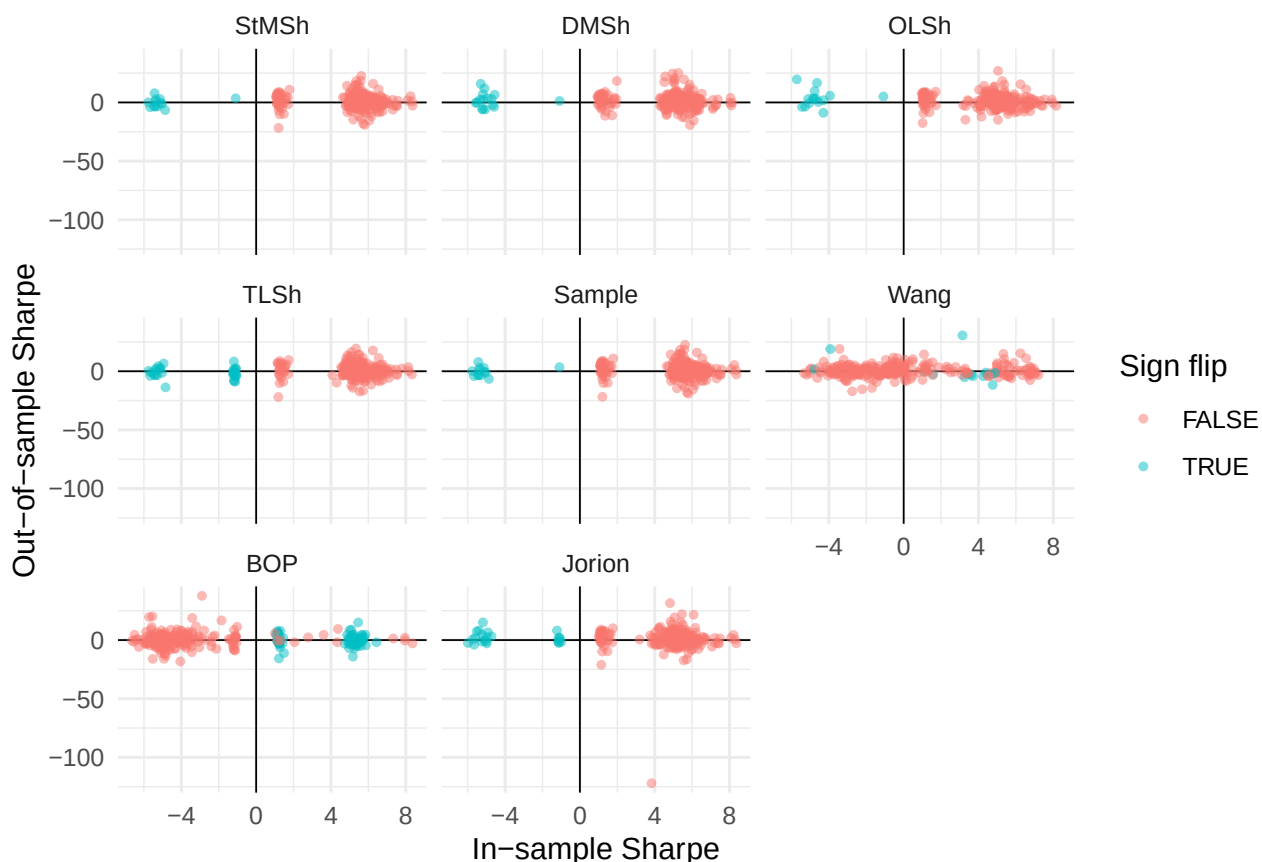


Figure 3: In-sample versus out-of-sample annualised Sharpe (tangency portfolios), by estimator. Net-short windows (sign flips) are highlighted.

1.6 The near-zero normaliser

Distinct from the sign flip, the tangency weights explode when c passes close to zero rather than merely reversing. Tangency gross leverage in window k is $\sum_i |z_i|/|c_k|$, so as $c_k \rightarrow 0$ the leverage diverges and every moment-based metric (return, volatility, certainty-equivalent, leverage, turnover) is dominated by that one window. The closest approach to zero across windows is smallest for **D-MSh**. At window 80 (30 January 2009), $c \approx 0.0002$ drives **TP_DMSH** gross leverage to 492,421. Table 10 lists the minimum $|c|$ per estimator.

Table 10: Closest approach of $c = \mathbf{1}^\top \Sigma^{-1} \hat{\mu}$ to zero, by estimator (smaller means a more unstable tangency window).

Estimator	Min $ c $
StMSh	0.0796
DMSH	0.0002
OLSh	0.4097
TLSh	0.0292
Sample	0.2447
Wang	0.2869
BOP	0.0219
Jorion	0.0162

1.7 Sample composition and the full universe

Figure 4 shows how our sample compares to the wider context of the full estimation universe. The top panel compares the equal-weighted cumulative return of the 100 Group 12 stocks against the full 441-stock universe, both re-based to 1 and shown on a log axis. The full-universe series is an equal-weighted proxy for the market rather than the cap-weighted S&P 500. The bottom panel decomposes the full universe into its GICS sectors on the same basis. The shaded bands mark the financial-crisis and pandemic sub-periods. The sector paths show why our sample's light Technology weighting muted its pandemic Sharpe ratios. The technology-led sectors that compounded fastest after 2020 are exactly the ones our draw underweights (Figure 1).

The full list of our 100 companies, grouped by GICS sector, is given in Table 11.

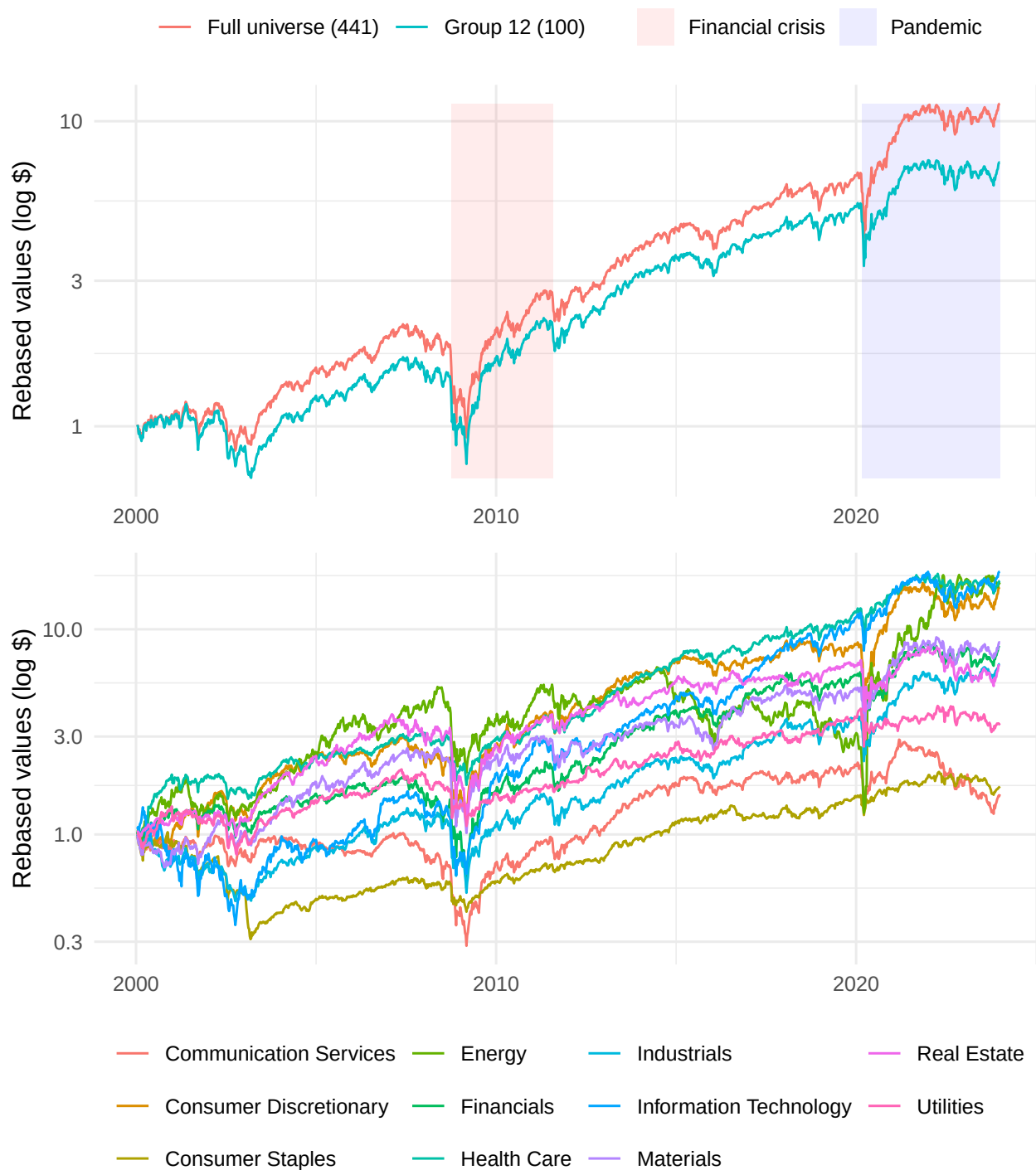


Figure 4: Rebased equal-weighted cumulative returns, January 2000 to December 2023, on a log axis. Top: the 100 Group 12 stocks against the full 441-stock universe. Bottom: the full universe split by GICS sector. Shaded bands mark the financial-crisis and pandemic sub-periods.

Table 11: Our (Group 12's) 100 sampled companies, grouped by GICS sector.

	Ticker	Company
Communication Services		
366	DISH	DISH NETWORK CORP
204	IPG	INTERPUBLIC GROUP COS INC
233	LUMN	LUMEN TECHNOLOGIES INC
127	OMC	OMNICOM GROUP INC
Consumer Discretionary		
39	FL	FOOT LOCKER INC
330	FOSL	FOSSIL GROUP INC
42	GT	GOODYEAR TIRE & RUBBER CO
215	JWN	NORDSTROM INC
236	LEG	LEGGETT & PLATT INC
239	LOW	LOWES COMPANIES INC
412	MAR	MARRIOTT INTERNATIONAL INC NEW
30	PVH	P V H CORP
399	RL	RALPH LAUREN CORP
440	ROST	ROSS STORES INC
194	SCI	SERVICE CORP INTERNATIONAL
154	TJX	T J X COMPANIES INC NEW
104	WHR	WHIRLPOOL CORP
Consumer Staples		
83	CHD	CHURCH & DWIGHT INC
430	COST	COSTCO WHOLESALE CORP NEW
103	CPB	CAMPBELLS CO
438	MNST	MONSTER BEVERAGE CORP NEW
184	TGT	TARGET CORP
Energy		
121	MUR	MURPHY OIL CORP
119	NC	NACCO INDUSTRIES INC
32	SLB	SCHLUMBERGER LTD
Financials		
216	AFL	AFLAC INC
266	AIG	AMERICAN INTERNATIONAL GROUP INC
336	ALL	ALLSTATE CORP
187	BK	BANK OF NEW YORK MELLON CORP
84	CINF	CINCINNATI FINANCIAL CORP
370	EG	EVEREST GROUP LTD
381	FDS	FACTSET RESEARCH SYSTEMS INC
141	FHN	FIRST HORIZON CORP
283	MBI	M B I A INC
180	MCO	MOODYS CORP

Table 11: Our (Group 12's) 100 sampled companies, grouped by GICS sector. *(continued)*

	Ticker	Company
269	MS	MORGAN STANLEY DEAN WITTER & CO
253	PGR	PROGRESSIVE CORP OH
284	SCHW	SCHWAB CHARLES CORP NEW
278	TFC	TRUIST FINANCIAL CORP
Health Care		
58	BMJ	BRISTOL MYERS SQUIBB CO
363	COR	CENCORA INC
391	DGX	QUEST DIAGNOSTICS INC
373	HSIC	SCHEIN HENRY INC
346	INCY	INCYTE CORP
72	PFE	PFIZER INC
313	PRGO	PERRIGO CO PLC
166	TFX	TELEFLEX INC
340	UHS	UNIVERSAL HEALTH SERVICES INC
376	WAT	WATERS CORP
247	WST	WEST PHARMACEUTICAL SERVICES INC
13	XRAY	DENTSPLY SIRONA INC
Industrials		
404	CHRW	CH ROBINSON WORLDWIDE INC
108	DOV	DOVER CORP
16	ETN	EATON CORP PLC
437	EXPD	EXPEDITORS INTERNATIONAL OF WASHINGTON INC
14	FAST	FASTENAL CO
201	GWV	GRAINGER W W INC
211	ITW	ILLINOIS TOOL WORKS INC
106	LHX	L3HARRIS TECHNOLOGIES INC
134	MAS	MASCO CORP
252	NSC	NORFOLK SOUTHERN CORP
140	ROL	ROLLINS INC
218	TEX	TEREX CORP NEW
409	URI	UNITED RENTALS INC
Information Technology		
34	AAPL	APPLE INC
152	DXC	D X C TECHNOLOGY CO
12	FICO	FAIR ISAAC CORP
75	GLW	CORNING INC
327	INTU	INTUIT INC
1	ORCL	ORACLE CORP
294	PTC	P T C INC
435	TDY	TELEDYNE TECHNOLOGIES

Table 11: Our (Group 12's) 100 sampled companies, grouped by GICS sector. *(continued)*

	Ticker	Company
300	TRMB	TRIMBLE INC
40	TXN	TEXAS INSTRUMENTS INC
298	TYL	TYLER TECHNOLOGIES INC
390	VYX	N C R VOYIX CORP
Materials		
167	AVY	AVERY DENNISON CORP
48	CCK	CROWN HOLDINGS INC
112	CLF	CLEVELAND CLIFFS INC NEW
275	ECL	ECOLAB INC
347	EMN	EASTMAN CHEMICAL CO
69	IP	INTERNATIONAL PAPER CO
67	NEM	NEWMONT CORP
136	NUE	NUCOR CORP
Real Estate		
219	FRT	FEDERAL REALTY INVESTMENT TR NEW
174	HST	HOST HOTELS & RESORTS INC
378	IRM	IRON MOUNTAIN INC NEW
342	REG	REGENCY CENTERS CORP
260	WELL	WELLTOWER INC
Utilities		
306	AES	A E S CORP
133	ATO	ATMOS ENERGY CORP
82	CMS	C M S ENERGY CORP
71	CNP	CENTERPOINT ENERGY INC
15	DTE	D T E ENERGY CO
116	DUK	DUKE ENERGY CORP NEW
165	ES	EVERSOURCE ENERGY
91	ETR	ENTERGY CORP NEW
88	PEG	PUBLIC SERVICE ENTERPRISE GP INC
77	PPL	P P L CORP
85	WEC	W E C ENERGY GROUP INC

2 Coursework 2: Supplementary analyses

2.1 Robustness checks

2.1.1 Sensitivity to the VehPower and BonusMalus caps

The preprocessing caps `VehPower` at 9 and `BonusMalus` at 150 before the log transform. Both cut-offs are varied here with all other choices held constant. The results are evaluated using the full-data Gamma GLM’s Akaike information criterion (AIC) alongside a single 5-fold CV of winsorised RMSE (95%), normalised Gini and the AE ratio.

Table 12: The effect of VehPower and BonusMalus cut-offs on the Gamma GLM (the values used were 9 and 150 respectively). AIC from the full-data fit; the remaining columns from a single 5-fold CV on a common fold partition.

Cap	AIC (full fit)	RMSE W95 (CV)	Gini norm (CV)	AE (CV)
VehPower cap (BonusMalus = 150)				
6	17040.0	1429	-0.142	1.058
9	17040.8	1429	-0.147	1.057
12	17039.9	1428	-0.142	1.057
none	17039.7	1429	-0.144	1.057
BonusMalus cap (VehPower = 9)				
100	17040.6	1431	-0.152	1.057
150	17040.8	1429	-0.147	1.057
200	17040.8	1429	-0.147	1.057
none	17040.8	1429	-0.147	1.057

As shown in Table 12, across all eight configurations the AIC spans barely a single unit (17,039.7 to 17,040.8), winsorised RMSE moves by 3 euros (€1428 to €1431), and the AE ratio is effectively constant (1.057 to 1.058). This shows that the defaults are neither favoured nor penalised and the model is insensitive to where these two variables are truncated. The `BonusMalus` cap matters less than the `VehPower` cap, with caps of 150, 200 and “none” resulting in identical figures. This is because no policy (in our 1,000 policies) carries a `BonusMalus` score above 150, so raising or removing the ceiling does not change the model. The only visible movement comes from the tighter cap of 100, which touches the 47 policies scoring above it and nudges RMSE up to 1431 (versus 1429 at the default). Scores above 100 are the malus-loaded, higher-risk policies, so cutting at 100 compresses real differentiation among the worst drivers. The normalised Gini stays negative throughout, consistent with the finding in B.1.iii. Policy-level severity carries no rankable out-of-sample signal at this sample size so its small movements across the grid are not informative.

2.1.2 Spline flexibility: three versus four degrees of freedom

Increasing the natural-spline degrees of freedom for `DrivAge` and `VehAge` from 3 to 4 adds one internal knot, and one parameter, per variable. As deviance alone fails to account for overfitting, we opted to primarily evaluate these models using AIC to penalise the additional parameters, paired with 5-fold cross-validation to assess true out-of-sample performance (Table 13).

Table 13: Comparison of spline flexibilities for driver and vehicle age, evaluated via full-data AIC, deviance, and 5-fold cross-validation metrics.

Degrees of freedom		AIC	Deviance	Params	RMSE W95 (CV)	Gini norm (CV)	AE (CV)
DrivAge	VehAge						
3	3	17040.8	1393.1	14	1429	-0.147	1.057
4	3	17042.8	1393.1	15	1436	-0.161	1.056
3	4	17032.2	1380.8	15	1418	-0.113	1.060
4	4	17034.1	1380.8	16	1425	-0.129	1.059

Adding a knot to `DrivAge` (while holding `VehAge` at 3) yields no benefit. The deviance remains completely unchanged at 1,393.1. As a result the AIC increases strictly due to the parameter penalty (from 17,040.8 to 17,042.8), and the winsorised RMSE worsens from 1429 to 1436. This indicates the extra driver-age flexibility is simply fitting noise.

Adding a knot to `VehAge` in isolation does improve the model. Deviance falls to 1,380.8, AIC decreases by roughly nine units (to 17,032.2), and winsorised RMSE improves to 1418. However this performance gain is modest (approximately a 0.7% improvement in winsorised RMSE) and serves better as an acknowledged limitation of the current specification rather than a mandate to re-run the full 500-run cross-validation study. The normalised Gini coefficient also remains negative across all configurations, showing that the increased flexibility provides no additional out-of-sample ranking power. We therefore retain three degrees of freedom for both variables, matching the specification of the released practice code and Asimit et al. [2025], and note the additional `VehAge` knot as a candidate refinement for a larger study.

2.1.3 Sensitivity to the target-encoding smoother κ

The credibility weight in the target encoding is defined as

$$\lambda = \frac{n_{\text{group}}}{n_{\text{group}} + \kappa}$$

with a default parameter of $\kappa = 20$. A smaller κ places more trust in sparse group means (risking overfitting), whereas a larger κ shrinks every group toward the global mean (risking underfitting). Because in-sample fit inherently rewards a smaller κ , we evaluate the comparison entirely out-of-sample.

As shown in Table 14, out-of-sample performance remains essentially flat across all values of κ with no single value performing materially better than the default. Winsorised RMSE varies by only a couple of euros across the entire grid (from 1428 at $\kappa = 5$ to 1430 at $\kappa = 100$), and the AE ratio is virtually unchanged (ranging from 1.057 to 1.058). Interestingly, a small κ does not lead to overfitting in this instance. The lowest winsorised RMSE occurs at the minimum tested value of $\kappa = 5$ suggesting that trusting sparse brand or region cells does not penalise the model at the current sample size. The only monotonic trend appears in the normalised Gini coefficient, which becomes steadily less negative as κ increases (from -0.153 to -0.130). However, since every recorded value remains negative, this

trend merely reflects the encoding being pulled more aggressively toward the global mean, rather than representing a genuine improvement in ranking power.

Table 14: Evaluating the Gamma GLM’s sensitivity to the target-encoding smoothing parameter κ . Performance is measured via a single 5-fold cross-validation on a shared partition, with 20 serving as the default value.

κ	RMSE W95 (CV)	Gini norm (CV)	AE (CV)
5	1428	-0.153	1.057
10	1429	-0.152	1.057
20	1429	-0.147	1.057
50	1429	-0.134	1.058
100	1430	-0.130	1.058

2.1.4 Standardisation

For `glm2`, `savvyGLM` and `glmnet` the fitted values depend on the data only through the linear predictor $x_i^\top \beta$. Scaling a column by s rescales its coefficient by $1/s$, leaving $\hat{\mu}_i$ unchanged. We note also that `glmnet` applies standardisation internally by default, ensuring that penalisation is applied on a common scale regardless of the input format.

To investigate this empirically, the full-data Gamma GLM was fitted with and without standardisation. The maximum absolute difference across all 1,000 fitted severities is approximately $5.0\text{e-}09$ (a negligible discrepancy attributable to floating-point rounding error). This confirms that standardisation in this context serves as a presentational convenience, retained to mirror the released practice code, rather than a substantive modelling choice that affects the results.

2.2 Convergence and computational notes

As an observational note, we encountered two types of warnings during the 500-run loop. Both originate from designed safeguards within the fitting algorithms rather than from data anomalies or implementation errors. Neither invalidates the reported results, but they are of technical interest and are described briefly below for completeness, alongside our interpretation.

glmnet: algorithm did not converge This appears when `glmnet` is asked to fit a Gamma model with a log link. For each candidate penalty strength, `glmnet` runs a small iterative fitting routine that is allowed up to 25 passes by default. With heavy-tailed severity data, the fits at the very lightly penalised end of the range occasionally need more than 25 passes, so they stop early and report it. This does not affect the final model. The penalty is chosen by cross-validation, and a fit that has not settled scores poorly and is passed over. We checked that the penalty finally selected never comes from that unsettled end of the range.

savvyGLM / glm.fit2: step size truncated / cannot correct step size The shrinkage GLMs use a fitting routine (`glm.fit2`) built around a simple safety check. At each step, if the proposed update would make the fit worse rather than better, the update is halved and tried again until it improves, and each time this happens a message is logged. This is common with a Gamma/log model when

the predictors are correlated. The “cannot correct step size” message means the routine used up its allowed halvings at that step and carried on from the best result it had so far. What actually matters is whether the fit settles in the end, and we record that for every fit. Across the 500 runs, the outcome is summarised in the table below. The number of passes stays far below the ceiling of 250, ranging from 6.3 for SR to 13.3 for GLM2, and every fit in this family (GLM2 and the six savvyGLM estimators) settles in all 500 runs. The glmnet models (RR and EN) do not expose a comparable per-fit flag, shown as a dash in the table, and their iteration counts measure coordinate-descent passes, so neither column is compared with the IRLS family.

Table 15: Convergence across the 500 evaluation runs. glmnet does not expose a per-fit convergence flag, shown as a dash. Its NoIt column counts coordinate-descent passes rather than IRLS iterations, so the two are not directly comparable.

Model	Converged (% of runs)	Mean NoIt
GLM2	100	13.3
RR	—	781.1
EN	—	1130.9
QIS	100	12.8
LW	100	11.6
SR	100	6.3
GSR	100	7.8
St	100	11.3
DSh	100	9.7

3 Interactive R Shiny dashboard

Since the rolling-window analysis in Part A depends on several choices that were fixed in the report (e.g., the group seed, the sample size, the training/testing window lengths, the risk-aversion parameter and the covariance estimator) a natural question is how sensitive the conclusions are to those choices. Rather than re-rendering the R Markdown document for each combination, we developed an interactive R Shiny dashboard that wraps the same estimator code (`All_eight_shrinkage_estimators.R`, `cov1Para.R`) and rolling-window engine used in the report, hosted on <https://www.shinyapps.io/>. The dashboard is available via:

- <https://3enji-apps.shinyapps.io/smm069-g12-part-a-dashboard/>

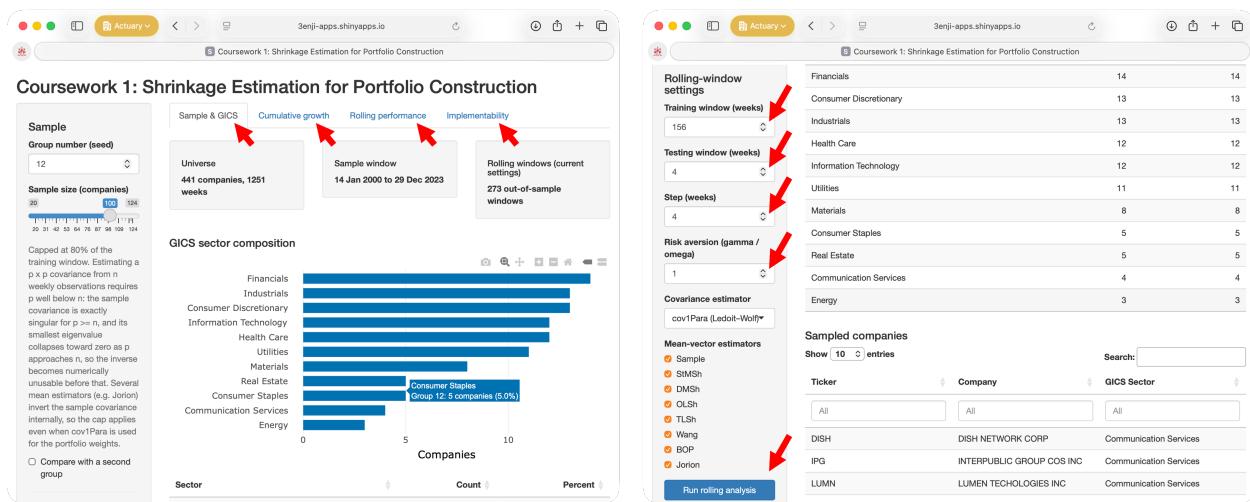


Figure 5: Our R Shiny dashboard. Left: the Sample & GICS tab, with the four analysis tabs highlighted. Right: the rolling-window controls (training, testing and step lengths, risk aversion, covariance estimator and mean-vector estimators) and the button that triggers the rolling analysis.

Figure 5 shows the R Shiny interface with the report's default settings (group 12, 100 companies). The dashboard is organised into four analysis tabs (Sample & GICS, Cumulative growth, Rolling performance, and Implementability). Each tab has the same sidebar of rolling-window controls: the training window, testing window and step (in weeks), the risk-aversion parameter γ , the covariance estimator (Ledoit–Wolf `cov1Para` or the sample covariance), and checkboxes to include or exclude any of the eight mean-vector estimators.

The sample-size slider is capped at 80% of the training window because several mean estimators (e.g. `Jorion`) invert the sample covariance internally. The sample covariance is exactly singular for $p \geq n$ and numerically unusable well before that, so the cap applies even when `cov1Para` supplies the portfolio-weight covariance. We also added a comparison switch that overlays a second group's stocks in the first two tabs.

The GICS and cumulative-growth tabs update instantly as the sample changes and the rolling analysis is gated behind a button. Figure 6 shows two of the interactive outputs. All plots are interactive and rendered with `plotly`. A strategy picker in the Rolling performance tab allows any subset of the mean–

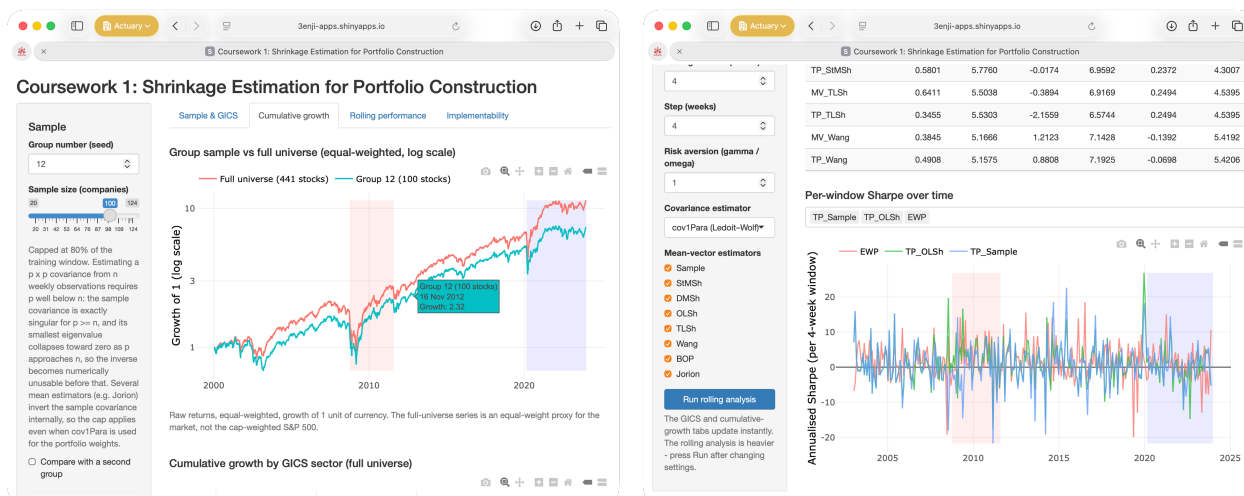


Figure 6: Left: the Cumulative growth tab comparing the group sample against the full universe on a log scale. Right: the Rolling performance tab with per-window annualised Sharpe ratios for a user-selected subset of strategies.

variance, tangency and equal-weighted portfolios to be compared window by window. A summary table reports the mean and standard deviation of per-window Sharpe ratios for the full period and both sub-periods.

Future extensions could include exporting run results, allowing for user-defined sub-period boundaries in place of the fixed Bai–Perron dates, and caching of completed runs so that different settings can be compared side by side without re-computation.

4 Reproducibility and generative AI declaration

4.1 Reproducibility & accessibility

An accessible HTML version of this report is available via a public GitHub page:

- https://github.com/ytterbiu/smm069-assignment_g12

This report was created in R Markdown. The source code is open source and is available via:

- https://ytterbiu.github.io/smm069-assignment_g12/

Changes made to this document were tracked using Git and are also available via the same repository (https://github.com/ytterbiu/smm069-assignment_g12/commits/main/). Please note that this commit history reflects updates made to the project repository and is not a comprehensive timeline of all contributions by group members.

4.2 Generative AI declaration

Generative AI (GenAI) tools were used throughout this project to assist with project planning, methodological validation, debugging, and the development of the supplementary interactive dashboard described in Appendix 3. In practice, these tools functioned as an additional technical resource, providing immediate feedback, troubleshooting code, and discussing conceptual assumptions.

Tools used

Google Gemini 3 Pro (Deep Research), Claude (Anthropic) Opus 4.8 & Fable 5.

Prompts used

The GenAI tools were provided with the coursework task sheet and extracts from our R code. A non-exhaustive list of prompts used is shown below. We opted to include at least one example of each type of prompt used, rather than a list of all prompts. For example, if we entered the prompts “How can we put two ggplots side by side?” and “How to remove grey background on a ggplot”, we would include only the first of these in the list below, as both relate to plotting issues (and are similar to querying with a traditional search engine).

- “...the top of both graphs in R are slightly different. (Pasting R code.) Can this be fixed for the plot area?”
- “Explain the repeated Sharpe-ratio values...”
- “Help to identify the key points here”
- “How can this error be tracked down in an Rmd file? ! Text line contains an invalid character. 1.1 ^^@^^@...”
- “Are there any issues with this app code? Suggest corrections for any potential bugs or errors within the server side code”

Use of GenAI outputs & subsequent revisions

The GenAI tools served as a consultative resource for the group. Notably, the AI output:

- Helped debug plotting issues and R Markdown compilation errors.
- Commented on the structure of R code snippets and the report structure.
- Assisted with converting draft content between document formats.
- Reviewed elements of the R Shiny app code, identifying bugs and design issues.

Changes made

All AI suggestions were critically reviewed and debated by the group before implementation. For example:

- While the AI suggested code for complex layouts, the specific aesthetic choices and annotations were manually adjusted to ensure clarity.
- Where the AI proposed specific interpretations, these were cross-referenced with academic literature (notably Asimit et al. [2026] and the lecture notes).
- In the case of the Shiny dashboard, the AI corrections were evaluated against the original code by the group. Each identified issue was assessed individually.
- The final R code for implementation was written and debugged by the group, with AI snippets being treated as drafts rather than final solutions.

While GenAI was treated as a collaborator, the final analysis, statistical interpretation, and the written content of this report were generated and verified by all group members.

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